LAWRENCE ONYANGO

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EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Bachelor of Science in Mechanical Engineering; Additional Major in Robotics; — *Class of 2024*

- GPA: 3.8 / 4.00; Graduated with University Honors
- **Relevant Coursework:** Optimal Control & Reinforcement Learning, Robotic Planning, Robot Kinematics & Dynamics, Computer Vision, Feedback Control Systems, Robotic Systems Engineering

SKILLS

Programming Languages: Python, C, C#, C++, Kotlin, MATLAB, Julia, Bash, Typescript, HTML, and CSS
Technologies: ROS, Kubernetes, Docker, SQL, NumPy, OpenCV, PyTorch and SolidWorks
Robotics: Kinematics, Dynamics, Computer Vision, Motion Planning, and Systems Design
Controls: Trajectory Optimization & Optimal Controls (MPC, LQR, DDP, DIRCOL) and Classical Controls (LTI Systems)

RESEARCH EXPERIENCE

University of Washington (SocialRL Lab) — **Seattle, WA**
Research Collaborator — *April 2025 - Present*

Human-AI Coordination Framework (DiZCo)

- Developed a human-AI coordination framework leveraging generative models for real-time planning and search-based trajectory selection, enabling agents to achieve zero-shot coordination with diverse partners.
- Implemented this framework by training a predictive, video-based world model conditioned on state, ego actions, and partner behavior, paired with a generative action proposer

PROFESSIONAL EXPERIENCE

Microsoft (Azure IoT Operations) — **Redmond, WA**
Software Engineer (Full Stack) — *July 2024 - Present*

- Contributed to the design and development of an in-house diagnostics platform, implementing a Go-based Kubernetes Operator with CRDs and Helm charts to automate lifecycle management of a remote diagnostics agent and enable declarative, scalable deployments.
- Developed a remote diagnostics agent deployed on customer clusters, enabling engineers to execute Kubernetes operations and collect logs securely, with command orchestration and session auditing for compliance.
- Built and maintained CI/CD pipelines to automate testing, integration, and deployment of diagnostics platform components, ensuring reliable delivery across environments.
- Helped design and develop an in-house web application for debugging Azure IoT Operations with React, TypeScript, Next.js, and .NET Framework, effectively reducing debugging time by up to 80% by centralizing and automating information display, eliminating manual data retrieval, and streamlining the troubleshooting process.
- Led critical offline testing efforts, collaborating with component teams to design comprehensive scenarios validating 72-hour durability in disconnected environments; uncovered and mitigated potential product gaps, preventing release blockers for General Availability launch.

Atlassian (Permissions Team) — **New York City, NY**
Software Engineer Intern — *May 2023 – Aug 2023*

- Consolidated and deployed two critical tier microservices utilizing Spring and Kotlin, streamlining sequence number generation and eliminating redundant code across multiple services. This unified approach removed the need for duplicated logic, resulting in a more efficient and maintainable system architecture.
- Designed customized dashboards using Splunk and Terraform, empowering individual service teams with dedicated monitoring tools for service performance.

Google (Cloud TI Platforms) — **Sunnyvale, CA**
Software Engineer Intern — *May 2022 – Aug 2022*

- Developed a remote diagnostic tool to access SSDs via SSH, automatically extract per-core logs, and perform detailed core-level analysis to identify and isolate malfunctioning units. Reduced debugging time for SSD developers by 30%, streamlining fault diagnosis and improving overall system reliability.

PUBLICATIONS

- **L. Onyango**, J. Yuan, Y. Du, N. Jaques. *DiZCo: Planning Zero-Shot Coordination in World Models*. Manuscript in preparation (initial version submitted to ICLR 2026)

PROJECTS

Autonomous Garden Robot *Fall 2023 – Spring 2024*

- Led development of a teleoperated garden robot integrating semi-autonomous navigation and intelligent plant care, performing automated soil sampling, targeted nutrient/water spraying, and real-time environmental data acquisition.
- Implemented navigation (Visual/LiDAR SLAM) and plant monitoring (stereo-camera) subsystems using ROS and engineered an AWS/Django web server for data management.

Safeguard Against Pests Robot *Fall 2023*

- Spearheaded development of an autonomous sentry robot for pest identification and elimination, designing and integrating a software stack with state machines and a YOLO-based computer vision system.
- Optimized the software stack, including the computer vision, to achieve real-time operation (sub-0.5 second response time) and implemented safety mechanisms to prevent unintended pesticide release.

5-DOF Jenga Building ARM *Fall 2023*

- Implemented classical robotics control for a 5-DOF robotic arm by deriving its Denavit–Hartenberg (DH) parameters and formulating both forward and inverse kinematics for precise end-effector control.
- Designed spline-based motion trajectories to achieve smooth, stable manipulation, enabling the robot to construct a six-layer Jenga tower with optimized position and velocity profiles.

Urban Search and Rescue Robot *Spring 2022*

- Built and manually operated a vision-guided LEGO robot with OpenCV & MATLAB to navigate a mock-disaster course with ramps, stairs, and fiducial markers, achieving one of the fastest completion times with all survivors rescued.